Arno Rüegg

Machine Learning Researcher · PhD candidate at ETH Zürich

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github.com/arrueegg 0009-0004-7410-458X 1 March 1999



Available from summer 2027 for applied ML roles in weather & climate, remote sensing, GNSS and geospatial analytics.

Profile

Machine-learning researcher who builds multi-modal models that fuse satellite and ground sensor data and know how uncertain they are. Over three years of taking heterogeneous observations to validated transformer and probabilistic deep-learning models on GPU clusters; four first-author papers, two of them peer-reviewed. Looking for applied problems where sensors are noisy and decisions need error bars.

Research Experience

Doctoral Researcher, *ETH Zürich, Space Geodesy group* 2023 – 2027 (expected)

- **Foundation model (Hemera).** Designing and training an 89M-parameter Perceiver-style transformer that fuses six observation types (GNSS, altimetry, radio occultation, ionosphere maps, space-weather indices) from 16 years into one latent global state; pretrained by predicting held-out observations, fine-tuned for 24 h forecasting.
- **Probabilistic modelling.** Bayesian residual MLP with per-observation uncertainty, pretrained on 11 years of GNSS data: **24.5% lower** single-frequency positioning error than the IGS maps, better on **96%** of test days.
- **Multi-sensor data fusion.** Weighted-loss fusion of dense GNSS with sparse VLBI data: **about 50% lower** median residuals near VLBI sites, validated on a year of independent Jason-3 altimetry.
- **Supervision.** Supervised **8 students** in four MSc projects and a BSc thesis (2024–2026), four of them on ML (GNNs, probabilistic networks, large-scale pretraining, LiDAR biomass mapping); a Data Science Lab team (ETH AI Center) I designed and supervised cut STEC test error by **22%** over its baseline with large-scale pretraining and solar indices.

Publications · all as first author

[1] Hemera: a Multi-Modal Foundation Model for the Global Ionospheric State. *in preparation*

[2] Probabilistic Machine Learning for Slant Total Electron Content Modelling based on GNSS. *JGR: MLC, submitted*

[3] Fusion of GNSS and VLBI Data for Global Ionospheric Mapping. *IAG Symposia (Springer), 2026*

[4] A Comparative Study of VTEC Estimates Derived from S/X VLBI and VGOS Observations. *Space Weather (AGU), 2025*

Education

PhD in Space Geodesy, *ETH Zürich* 2023 – 2027 (expected)

MSc Geomatics (Geoinformatics), *ETH Zürich* 2021 – 2023

- Thesis: *Shared Deep Feature Embedding of Sentinel-1 and Sentinel-2 for Building Detection*. SAR–optical segmentation with adversarial domain alignment; **+3.1 pts IoU** from Stable Diffusion VAE features.
- Project: *Vegetation Mapping with ICESat-2*. Bias-corrected a canopy-height map: RMSE **7.6** → **6.0 m** in seven regions.

BSc Geospatial Engineering, *ETH Zürich* 2018 – 2021

- Thesis: *Classification of Solid and Loose Rock on Orthophotos with Deep Learning*. U-Net, **84.5%** test accuracy.

Skills

Machine learning	Foundation models, transformers, self-supervised pretraining, multi-modal data fusion, probabilistic deep learning, uncertainty quantification, spatiotemporal forecasting, computer vision
Engineering	Python, PyTorch (bf16 mixed precision), NumPy, pandas, SciPy, pytest, W&B, Bash, Git, AI coding agents (Claude Code); basic C++, MATLAB, SQL
Data & compute	xarray, Zarr, HDF5, netCDF, Parquet; Matplotlib, Cartopy; SLURM on ETH's Euler HPC cluster (A100 GPUs)
Domain	Ionosphere & space weather, GNSS, VLBI, Earth observation, parameter estimation, photogrammetry

Leadership & More

Leadership	Head of scout department, Pfadi Trogen: led four teams of 6–8 volunteer leaders	2021 – 2025
Teaching	Teaching assistant: Geodetic Data Analysis, Measurement Techniques, Linear Algebra	2021 – 2026
Languages	German (native), English (fluent), French (basic)	
Interests	Hiking, skiing, football, scouting	